

# IS42/45R86400D/16320D/32160D IS42/45S86400D/16320D/32160D



**16Mx32, 32Mx16, 64Mx8  
512Mb SDRAM**

**PRELIMINARY INFORMATION  
JUNE 2011**

## FEATURES

- Clock frequency: 200, 166, 143 MHz
- Fully synchronous; all signals referenced to a positive clock edge
- Internal bank for hiding row access/precharge
- Power supply:  $V_{DD}/V_{DDQ} = 2.3V-3.6V$   
IS42/45SxxxxxD -  $V_{DD}/V_{DDQ} = 3.3V$   
IS42/45RxxxxxD -  $V_{DD}/V_{DDQ} = 2.5$
- LVTTTL interface
- Programmable burst length  
– (1, 2, 4, 8, full page)
- Programmable burst sequence:  
Sequential/Interleave
- Auto Refresh (CBR)
- Self Refresh
- 8K refresh cycles every 64 ms
- Random column address every clock cycle
- Programmable CAS latency (2, 3 clocks)
- Burst read/write and burst read/single write operations capability
- Burst termination by burst stop and precharge command
- Packages:  
x8/x16: 54-pin TF-TSOP-II, 54-ball TF-BGA (x16 only)  
x32: 90-ball TF-BGA
- Temperature Range:  
Commercial (0°C to +70°C)  
Industrial (-40°C to +85°C)  
Automotive, A1 (-40°C to +85°C)  
Automotive, A2 (-40°C to +105°C)

## DEVICE OVERVIEW

ISSI's 512Mb Synchronous DRAM achieves high-speed data transfer using pipeline architecture. All inputs and outputs signals refer to the rising edge of the clock input. The 512Mb SDRAM is organized as follows.

## PACKAGE INFORMATION

| IS42/45S32160D       | IS42/45S16320D                   | IS42/45S86400D       |
|----------------------|----------------------------------|----------------------|
| IS42/45R32160D       | IS42/45R16320D                   | IS42/45R86400D       |
| 4M x 32 x 4<br>banks | 8M x 16 x 4<br>banks             | 16M x 8 x 4<br>banks |
| 90-ball TF-BGA       | 54-pin TSOP-II<br>54-ball TF-BGA | 54-pin TSOP-II       |

## KEY TIMING PARAMETERS

| Parameter              | -5  | -6  | -7  | Unit |
|------------------------|-----|-----|-----|------|
| Clk Cycle Time         |     |     |     |      |
| CAS Latency = 3        | 5   | 6   | 7   | ns   |
| CAS Latency = 2        | 10  | 10  | 7.5 | ns   |
| Clk Frequency          |     |     |     |      |
| CAS Latency = 3        | 200 | 167 | 143 | Mhz  |
| CAS Latency = 2        | 100 | 100 | 133 | Mhz  |
| Access Time from Clock |     |     |     |      |
| CAS Latency = 3        | 5.0 | 5.4 | 5.4 | ns   |
| CAS Latency = 2        | 6   | 6   | 5.4 | ns   |

## ADDRESS TABLE

| Parameter             | 16M x 32             | 32M x 16             | 64M x 8              |
|-----------------------|----------------------|----------------------|----------------------|
| Configuration         | 4M x 32 x 4<br>banks | 8M x 16 x 4<br>banks | 16M x 8 x 4<br>banks |
| Bank Address<br>Pins  | BA0, BA1             | BA0, BA1             | BA0, BA1             |
| Autoprecharge<br>Pins | A10/AP               | A10/AP               | A10/AP               |
| Row Address           | 8K(A0 – A12)         | 8K(A0 – A12)         | 8K(A0 – A12)         |
| Column<br>Address     | 512(A0 – A8)         | 1K(A0 – A9)          | 2K(A0 – A9,<br>A11)  |
| Refresh Count         |                      |                      |                      |
| Com./Ind./A1          | 8K / 64ms            | 8K / 64ms            | 8K / 64ms            |
| A2                    | 8K / 16ms            | 8K / 16ms            | 8K / 16ms            |

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## PIN FUNCTIONS

| Symbol                                        | Type             | Function (In Detail)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A0-A12                                        | Input Pin        | Address Inputs: A0-A12 are sampled during the ACTIVE command (row-address A0-A12) and READ/WRITE command (column address A0-A9, A11 (x8); A0-A9 (x16); A0-A8 (x32); with A10 defining auto precharge) to select one location out of the memory array in the respective bank. A10 is sampled during a PRECHARGE command to determine if all banks are to be precharged (A10 HIGH) or bank selected by BA0, BA1 (LOW). The address inputs also provide the op-code during a LOAD MODE REGISTER command.                                                                                                                         |
| BA0, BA1                                      | Input Pin        | Bank Select Address: BA0 and BA1 defines which bank the ACTIVE, READ, WRITE or PRECHARGE command is being applied.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| $\overline{\text{CAS}}$                       | Input Pin        | $\overline{\text{CAS}}$ , in conjunction with the $\overline{\text{RAS}}$ and $\overline{\text{WE}}$ , forms the device command. See the "Command Truth Table" for details on device commands.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| CKE                                           | Input Pin        | The CKE input determines whether the CLK input is enabled. The next rising edge of the CLK signal will be valid when is CKE HIGH and invalid when LOW. When CKE is LOW, the device will be in either power-down mode, clock suspend mode, or self refresh mode. CKE is an asynchronous input.                                                                                                                                                                                                                                                                                                                                 |
| CLK                                           | Input Pin        | CLK is the master clock input for this device. Except for CKE, all inputs to this device are acquired in synchronization with the rising edge of this pin.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $\overline{\text{CS}}$                        | Input Pin        | The $\overline{\text{CS}}$ input determines whether command input is enabled within the device. Command input is enabled when $\overline{\text{CS}}$ is LOW, and disabled with $\overline{\text{CS}}$ is HIGH. The device remains in the previous state when $\overline{\text{CS}}$ is HIGH.                                                                                                                                                                                                                                                                                                                                  |
| DQM: x8<br>DQML, DQMH: x16<br>DQM0-DQM3: x32  | Input Pin        | DQx pins control the bytes of the I/O buffers. For example with x16, in read mode, DQML and DQMH control the output buffer. When DQML or DQMH is LOW, the corresponding buffer byte is enabled, and when HIGH, disabled. The outputs go to the HIGH impedance state when DQML/DQMH is HIGH. This function corresponds to $\overline{\text{OE}}$ in conventional DRAMs. In write mode, DQML and DQMH control the input buffer. When DQML or DQMH is LOW, the corresponding buffer byte is enabled, and data can be written to the device. When DQML or DQMH is HIGH, input data is masked and cannot be written to the device. |
| DQ0-DQ7: x8<br>DQ0-DQ15: x16<br>DQ0-DQ31: x32 | Input/Output     | Data on the Data Bus is latched on DQ pins during Write commands, and buffered for output after Read commands.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| $\overline{\text{RAS}}$                       | Input Pin        | $\overline{\text{RAS}}$ , in conjunction with $\overline{\text{CAS}}$ and $\overline{\text{WE}}$ , forms the device command. See the "Command Truth Table" item for details on device commands.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| $\overline{\text{WE}}$                        | Input Pin        | $\overline{\text{WE}}$ , in conjunction with $\overline{\text{RAS}}$ and $\overline{\text{CAS}}$ , forms the device command. See the "Command Truth Table" item for details on device commands.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| VDDQ                                          | Power Supply Pin | VDDQ is the output buffer power supply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| VDD                                           | Power Supply Pin | VDD is the device internal power supply.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| VSSQ                                          | Power Supply Pin | VSSQ is the output buffer ground.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| VSS                                           | Power Supply Pin | VSS is the device internal ground.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## GENERAL DESCRIPTION

### READ

The READ command selects the bank from BA0, BA1 inputs and starts a burst read access to an active row. Inputs A0-An (For column addresses, n=A8 for x32, n=A9 for x16, n=A11 for x8), provides the starting column location. When A10 is HIGH, this command functions as an AUTO PRECHARGE command. When the auto precharge is selected, the row being accessed will be precharged at the end of the READ burst. The row will remain open for subsequent accesses when AUTO PRECHARGE is not selected. DQ's read data is subject to the logic level on the DQM inputs two clocks earlier. When a given DQM signal was registered HIGH, the corresponding DQ's will be High-Z two clocks later. DQ's will provide valid data when the DQM signal was registered LOW.

### WRITE

A burst write access to an active row is initiated with the WRITE command. BA0, BA1 inputs selects the bank, and the starting column location is provided by inputs A0-An (For column addresses, n=A8 for x32, n=A9 for x16, n=A11 for x8). AUTO-PRECHARGE is determined by A10.

The row being accessed will be precharged at the end of the WRITE burst, if AUTO PRECHARGE is selected. If AUTO PRECHARGE is not selected, the row will remain open for subsequent accesses.

A memory array is written with corresponding input data on DQ's and DQM input logic level appearing at the same time. Data will be written to memory when DQM signal is LOW. When DQM is HIGH, the corresponding data inputs will be ignored, and a WRITE will not be executed to that byte/column location.

### PRECHARGE

The PRECHARGE command is used to deactivate the open row in a particular bank or the open row in all banks. BA0, BA1 can be used to select which bank is precharged or they are treated as "Don't Care". A10 determined whether one or all banks are precharged. After executing this command, the next command for the selected bank(s) is executed after passage of the period  $t_{RP}$ , which is the period required for bank precharging. Once a bank has been precharged, it is in the idle state and must be activated prior to any READ or WRITE commands being issued to that bank.

### AUTO PRECHARGE

The AUTO PRECHARGE function ensures that the precharge is initiated at the earliest valid stage within a burst. This function allows for individual-bank precharge without

requiring an explicit command. A10 to enable the AUTO PRECHARGE function in conjunction with a specific READ or WRITE command. For each individual READ or WRITE command, auto precharge is either enabled or disabled. AUTO PRECHARGE does not apply except in full-page burst mode. Upon completion of the READ or WRITE burst, a precharge of the bank/row that is addressed is automatically performed.

### AUTO REFRESH COMMAND

This command executes the AUTO REFRESH operation. The row address and bank to be refreshed are automatically generated during this operation. The stipulated period ( $t_{RC}$ ) is required for a single refresh operation, and no other commands can be executed during this period. This command is executed at least 8192 times for every  $T_{REF}$  period. During an AUTO REFRESH command, address bits are "Don't Care". This command corresponds to CBR Auto-refresh.

### BURST TERMINATE

The BURST TERMINATE command forcibly terminates the burst read and write operations by truncating either fixed-length or full-page bursts and the most recently registered READ or WRITE command prior to the BURST TERMINATE.

### COMMAND INHIBIT

COMMAND INHIBIT prevents new commands from being executed. Operations in progress are not affected, apart from whether the CLK signal is enabled

### NO OPERATION

When  $\overline{CS}$  is low, the NOP command prevents unwanted commands from being registered during idle or wait states.

### LOAD MODE REGISTER

During the LOAD MODE REGISTER command the mode register is loaded from A0-A12. This command can only be issued when all banks are idle.

### ACTIVE COMMAND

When the ACTIVE COMMAND is activated, BA0, BA1 inputs selects a bank to be accessed, and the address inputs on A0-A12 selects the row. Until a PRECHARGE command is issued to the bank, the row remains open for accesses.

**COMMAND TRUTH TABLE**

| Function                    | CKE   |   | $\overline{CS}$ | $\overline{RAS}$ | $\overline{CAS}$ | $\overline{WE}$ | BA1 | BA0 | A12, A11 |         |
|-----------------------------|-------|---|-----------------|------------------|------------------|-----------------|-----|-----|----------|---------|
|                             | n - 1 | n |                 |                  |                  |                 |     |     | A10      | A9 - A0 |
| Device deselect (DESL)      | H     | x | H               | x                | x                | x               | x   | x   | x        | x       |
| No operation (NOP)          | H     | x | L               | H                | H                | H               | x   | x   | x        | x       |
| Burst stop (BST)            | H     | x | L               | H                | H                | L               | x   | x   | x        | x       |
| Read                        | H     | x | L               | H                | L                | H               | V   | V   | L        | V       |
| Read with auto precharge    | H     | x | L               | H                | L                | H               | V   | V   | H        | V       |
| Write                       | H     | x | L               | H                | L                | L               | V   | V   | L        | V       |
| Write with auto precharge   | H     | x | L               | H                | L                | L               | V   | V   | H        | V       |
| Bank activate (ACT)         | H     | x | L               | L                | H                | H               | V   | V   | V        | V       |
| Precharge select bank (PRE) | H     | x | L               | L                | H                | L               | V   | V   | L        | x       |
| Precharge all banks (PALL)  | H     | x | L               | L                | H                | L               | x   | x   | H        | x       |
| CBR Auto-Refresh (REF)      | H     | H | L               | L                | L                | H               | x   | x   | x        | x       |
| Self-Refresh (SELF)         | H     | L | L               | L                | L                | H               | x   | x   | x        | x       |
| Mode register set (MRS)     | H     | x | L               | L                | L                | L               | L   | L   | L        | V       |

Note: H=V<sub>IH</sub>, L=V<sub>IL</sub> x= V<sub>IH</sub> or V<sub>IL</sub>, V = Valid Data.

**DQM TRUTH TABLE**

| Function                                  | CKE |   | DQMH | DQML |
|-------------------------------------------|-----|---|------|------|
|                                           | n-1 | n |      |      |
| Data write / output enable                | H   | x | L    | L    |
| Data mask / output disable                | H   | x | H    | H    |
| Upper byte write enable / output enable   | H   | x | L    | x    |
| Lower byte write enable / output enable   | H   | x | x    | L    |
| Upper byte write inhibit / output disable | H   | x | H    | x    |
| Lower byte write inhibit / output disable | H   | x | x    | H    |

Note:

1. H=V<sub>IH</sub>, L=V<sub>IL</sub> x= V<sub>IH</sub> or V<sub>IL</sub>, V = Valid Data.

2. x16 options shown. x32 DQM0-DQM3 is similar in function.

**AC ELECTRICAL CHARACTERISTICS** (1,2,3)

| Symbol           | Parameter                                                                                                                                                           |                                        | -5   |      | -6   |      | -7   |      | Units |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|------|------|------|------|------|------|-------|
|                  |                                                                                                                                                                     |                                        | Min. | Max. | Min. | Max. | Min. | Max. |       |
| t <sub>CK3</sub> | Clock Cycle Time                                                                                                                                                    | $\overline{\text{CAS}}$ Latency = 3    | 5    | —    | 6    | —    | 7    | —    | ns    |
| t <sub>CK2</sub> |                                                                                                                                                                     | $\overline{\text{CAS}}$ Latency = 2    | 10   | —    | 10   | —    | 7.5  | —    | ns    |
| t <sub>AC3</sub> | Access Time From CLK                                                                                                                                                | $\overline{\text{CAS}}$ Latency = 3    | —    | 5.0  | —    | 5.4  | —    | 5.4  | ns    |
| t <sub>AC2</sub> |                                                                                                                                                                     | $\overline{\text{CAS}}$ Latency = 2    | —    | 6    | —    | 6    | —    | 5.4  | ns    |
| t <sub>CH</sub>  | CLK HIGH Level Width                                                                                                                                                |                                        | 2    | —    | 2.5  | —    | 2.5  | —    | ns    |
| t <sub>CL</sub>  | CLK LOW Level Width                                                                                                                                                 |                                        | 2    | —    | 2.5  | —    | 2.5  | —    | ns    |
| t <sub>OH3</sub> | Output Data Hold Time                                                                                                                                               | $\overline{\text{CAS}}$ Latency = 3    | 2.5  | —    | 2.7  | —    | 2.7  | —    | ns    |
| t <sub>OH2</sub> |                                                                                                                                                                     | $\overline{\text{CAS}}$ Latency = 2    | 2.5  | —    | 2.7  | —    | 2.7  | —    | ns    |
| t <sub>LZ</sub>  | Output LOW Impedance Time                                                                                                                                           |                                        | 0    | —    | 0    | —    | 0    | —    | ns    |
| t <sub>HZ3</sub> | Output HIGH Impedance Time                                                                                                                                          |                                        | —    | 5    | —    | 5.4  | —    | 5.4  | ns    |
| t <sub>HZ2</sub> |                                                                                                                                                                     |                                        | —    | 6    | —    | 6    | —    | 5.4  | ns    |
| t <sub>DS</sub>  | Input Data Setup Time <sup>(2)</sup>                                                                                                                                |                                        | 1.8  | —    | 1.5  | —    | 1.5  | —    | ns    |
| t <sub>DH</sub>  | Input Data Hold Time <sup>(2)</sup>                                                                                                                                 |                                        | 0.8  | —    | 0.8  | —    | 0.8  | —    | ns    |
| t <sub>AS</sub>  | Address Setup Time <sup>(2)</sup>                                                                                                                                   |                                        | 1.5  | —    | 1.5  | —    | 1.5  | —    | ns    |
| t <sub>AH</sub>  | Address Hold Time <sup>(2)</sup>                                                                                                                                    |                                        | 0.8  | —    | 0.8  | —    | 0.8  | —    | ns    |
| t <sub>CKS</sub> | CKE Setup Time <sup>(2)</sup>                                                                                                                                       |                                        | 1.5  | —    | 1.5  | —    | 1.5  | —    | ns    |
| t <sub>CKH</sub> | CKE Hold Time <sup>(2)</sup>                                                                                                                                        |                                        | 0.8  | —    | 0.8  | —    | 0.8  | —    | ns    |
| t <sub>CMS</sub> | Command Setup Time ( $\overline{\text{CS}}$ , $\overline{\text{RAS}}$ , $\overline{\text{CAS}}$ , $\overline{\text{WE}}$ , $\overline{\text{DQM}}$ ) <sup>(2)</sup> |                                        | 1.5  | —    | 1.5  | —    | 1.5  | —    | ns    |
| t <sub>CMH</sub> | Command Hold Time ( $\overline{\text{CS}}$ , $\overline{\text{RAS}}$ , $\overline{\text{CAS}}$ , $\overline{\text{WE}}$ , $\overline{\text{DQM}}$ ) <sup>(2)</sup>  |                                        | 0.8  | —    | 0.8  | —    | 0.8  | —    | ns    |
| t <sub>RC</sub>  | Command Period (REF to REF / ACT to ACT)                                                                                                                            |                                        | 55   | —    | 60   | —    | 60   | —    | ns    |
| t <sub>RAS</sub> | Command Period (ACT to PRE)                                                                                                                                         |                                        | 38   | 100K | 42   | 100K | 37   | 100K | ns    |
| t <sub>RP</sub>  | Command Period (PRE to ACT)                                                                                                                                         |                                        | 15   | —    | 18   | —    | 15   | —    | ns    |
| t <sub>RCD</sub> | Active Command To Read / Write Command Delay Time                                                                                                                   |                                        | 15   | —    | 18   | —    | 15   | —    | ns    |
| t <sub>RRD</sub> | Command Period (ACT [0] to ACT[1])                                                                                                                                  |                                        | 10   | —    | 12   | —    | 14   | —    | ns    |
| t <sub>DPL</sub> | Input Data To Precharge Command Delay time                                                                                                                          |                                        | 10   | —    | 12   | —    | 14   | —    | ns    |
| t <sub>DAL</sub> | Input Data To Active / Refresh Command Delay time (During Auto-Precharge)                                                                                           |                                        | 25   | —    | 30   | —    | 29   | —    | ns    |
| t <sub>MRD</sub> | Mode Register Program Time                                                                                                                                          |                                        | 10   | —    | 12   | —    | 14   | —    | ns    |
| t <sub>DDE</sub> | Power Down Exit Setup Time                                                                                                                                          |                                        | 5    | —    | 6    | —    | 7    | —    | ns    |
| t <sub>XS</sub>  | Self-Refresh Exit Time                                                                                                                                              |                                        | 60   | —    | 70   | —    | 67   | —    | ns    |
| t <sub>T</sub>   | Transition Time                                                                                                                                                     |                                        | 0.3  | 1.2  | 0.3  | 1.2  | 0.3  | 1.2  | ns    |
| t <sub>REF</sub> | Refresh Cycle Time (8192)                                                                                                                                           | T <sub>A</sub> ≤ 70°C Com, Ind, A1, A2 | —    | 64   | —    | 64   | —    | 64   | ms    |
|                  |                                                                                                                                                                     | T <sub>A</sub> ≤ 85°C Ind, A1, A2      | —    | 64   | —    | 64   | —    | 64   | ms    |
|                  |                                                                                                                                                                     | T <sub>A</sub> > 85°C A2               | —    | —    | —    | —    | —    | 16   | ms    |

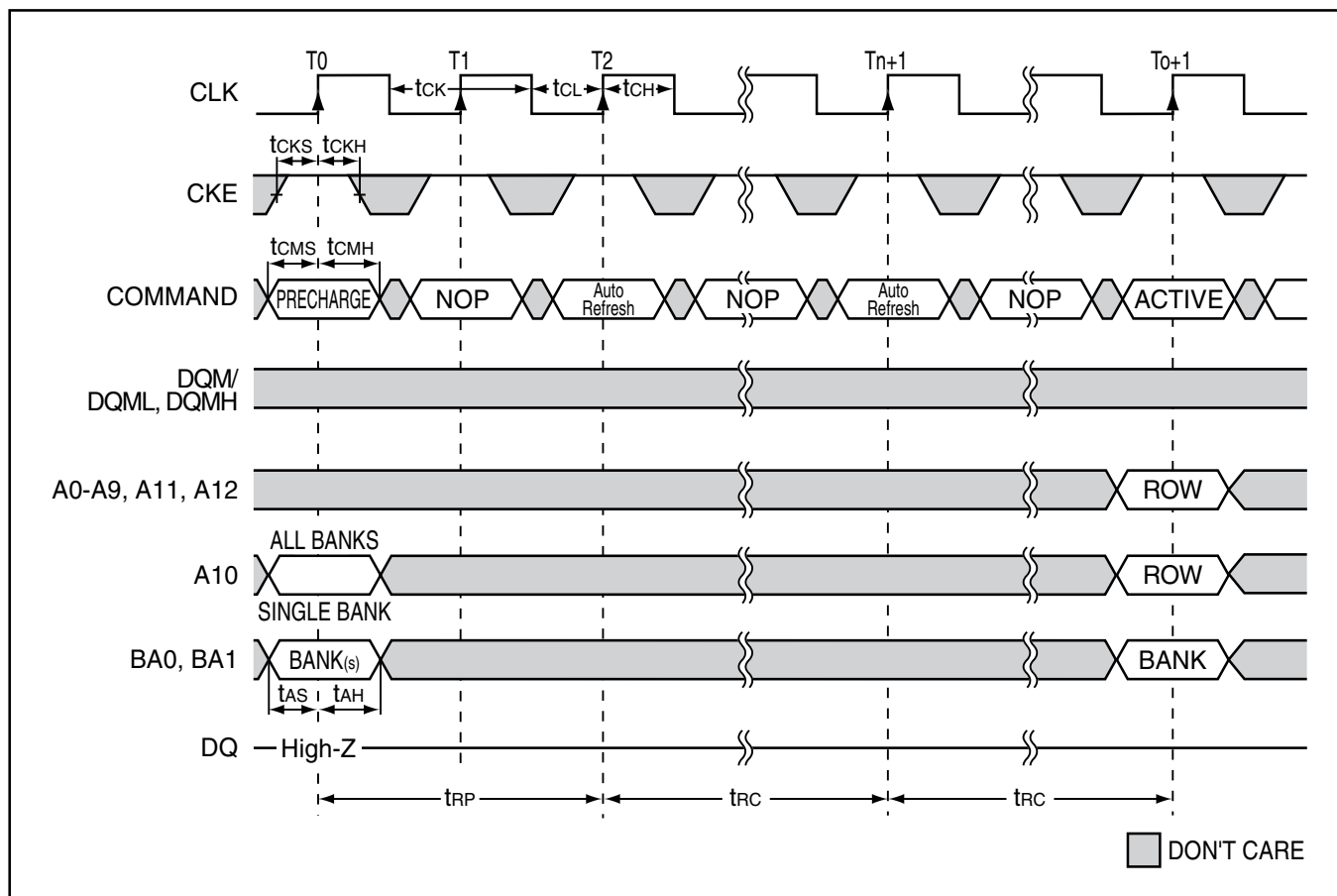
**Notes:**

1. The power-on sequence must be executed before starting memory operation.
2. Measured with t<sub>T</sub> = 1 ns. If clock rising time is longer than 1ns, (t<sub>T</sub> / 2 - 0.5) ns should be added to the parameter.
3. The reference level is 1.4V when measuring input signal timing. Rise and fall times are measured between V<sub>IH</sub>(min.) and V<sub>IL</sub>(max).

## OPERATING FREQUENCY / LATENCY RELATIONSHIPS

| SYMBOL           | PARAMETER                                                               | -5                                  | -6  | -7  | -7  | UNITS |
|------------------|-------------------------------------------------------------------------|-------------------------------------|-----|-----|-----|-------|
| —                | Clock Cycle Time                                                        | 5                                   | 6   | 7   | 7.5 | ns    |
| —                | Operating Frequency                                                     | 200                                 | 167 | 143 | 133 | MHz   |
| t <sub>CAC</sub> | $\overline{\text{CAS}}$ Latency                                         | 3                                   | 3   | 3   | 2   | cycle |
| t <sub>RCD</sub> | Active Command To Read/Write Command Delay Time                         | 3                                   | 3   | 3   | 2   | cycle |
| t <sub>RAC</sub> | $\overline{\text{RAS}}$ Latency (t <sub>RCD</sub> + t <sub>CAC</sub> )  | $\overline{\text{CAS}}$ Latency = 3 | 6   | 6   | 6   | —     |
|                  |                                                                         | $\overline{\text{CAS}}$ Latency = 2 | —   | —   | —   | 4     |
| t <sub>RC</sub>  | Command Period (REF to REF / ACT to ACT)                                | 10                                  | 10  | 9   | 8   | cycle |
| t <sub>RAS</sub> | Command Period (ACT to PRE)                                             | 7                                   | 7   | 6   | 5   | cycle |
| t <sub>RP</sub>  | Command Period (PRE to ACT)                                             | 3                                   | 3   | 3   | 2   | cycle |
| t <sub>RRD</sub> | Command Period (ACT[0] to ACT [1])                                      | 2                                   | 2   | 2   | 2   | cycle |
| t <sub>CCD</sub> | Column Command Delay Time (READ, READA, WRIT, WRITA)                    | 1                                   | 1   | 1   | 1   | cycle |
| t <sub>DPL</sub> | Input Data To Precharge Command Delay Time                              | 2                                   | 2   | 2   | 2   | cycle |
| t <sub>DAL</sub> | Input Data To Active/Refresh Command Delay Time (During Auto-Precharge) | 5                                   | 5   | 5   | 4   | cycle |
| t <sub>RBD</sub> | Burst Stop Command To Output in HIGH-Z Delay Time (Read)                | $\overline{\text{CAS}}$ Latency = 3 | 3   | 3   | 3   | —     |
|                  |                                                                         | $\overline{\text{CAS}}$ Latency = 2 | —   | —   | —   | 2     |
| t <sub>WBD</sub> | Burst Stop Command To Input in Invalid Delay Time (Write)               | 0                                   | 0   | 0   | 0   | cycle |
| t <sub>RQL</sub> | Precharge Command To Output in HIGH-Z Delay Time (Read)                 | $\overline{\text{CAS}}$ Latency = 3 | 3   | 3   | 3   | —     |
|                  |                                                                         | $\overline{\text{CAS}}$ Latency = 2 | —   | —   | —   | 2     |
| t <sub>WDL</sub> | Precharge Command To Input in Invalid Delay Time (Write)                | 0                                   | 0   | 0   | 0   | cycle |
| t <sub>PQL</sub> | Last Output To Auto-Precharge Start Time (Read)                         | $\overline{\text{CAS}}$ Latency = 3 | -2  | -2  | -2  | —     |
|                  |                                                                         | $\overline{\text{CAS}}$ Latency = 2 | —   | —   | —   | -1    |
| t <sub>QMD</sub> | DQM To Output Delay Time (Read)                                         | 2                                   | 2   | 2   | 2   | cycle |
| t <sub>DMD</sub> | DQM To Input Delay Time (Write)                                         | 0                                   | 0   | 0   | 0   | cycle |
| t <sub>MRD</sub> | Mode Register Set To Command Delay Time                                 | 2                                   | 2   | 2   | 2   | cycle |

## AUTO-REFRESH CYCLE



### Notes:

1. CAS latency = 2, 3